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科目：概率論與隨機過程

1. Solution: Denote the events

 $A = \{\text{one face is head}\}$ $B = \{\text{one face is opposite face is tail}\}$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{1}{3}}{\frac{2}{3}} = \frac{1}{2}$$

2. Proof: ~~Denote that the ball chosen~~Denote ~~that~~ the possibility of the ball chosen from jar k is white is $P_{w,k}$

Hence

$$P_{w,k} = P_{w,k-1} \cdot \frac{m+1}{m+n+1} + (1 - P_{w,k-1}) \cdot \frac{m}{m+n+1}$$

$$= \frac{m}{m+n+1} + \frac{1}{m+n+1} P_{w,k-1}$$

$$\Leftrightarrow \left(P_{w,k} - \frac{m}{m+n} \right) = \frac{1}{m+n+1} \left(P_{w,k-1} - \frac{m}{m+n} \right)$$

$$\Rightarrow P_{w,k} - \frac{m}{m+n} = \left(\frac{1}{m+n+1} \right)^{k-1} \left(P_{w,1} - \frac{m}{m+n} \right)$$

$$\text{Obviously, } P_{w,1} = \frac{m}{m+n} \Rightarrow P_{w,k} - \frac{m}{m+n} = \left(\frac{1}{m+n+1} \right)^{k-1} \cdot 0 = 0$$

$$\Leftrightarrow P_{w,k} = \frac{m}{m+n}$$

$$\begin{aligned}
 3. (a) P(\text{a total of 4 games}) &= P(A \text{ wins 1st game} \cap B \text{ wins 2nd, 3rd, 4th games}) \\
 &\quad + P(B \text{ wins 1st game} \cap A \text{ wins 2nd, 3rd, 4th games}) \\
 &= p(1-p)^3 + (1-p)p^3
 \end{aligned}$$

(b) Denote $P(A \text{ wins}) = Q$

~~$Q = P(A \text{ wins 1st game})$~~

$$Q = p^2 + p(1-p)Q + (1-p)pQ \Leftrightarrow (2p^2 - 2p + 1)Q = p^2$$

$$\Leftrightarrow Q = \frac{p^2}{p^2 + (p-1)^2} = P(A \text{ wins})$$

$P(A \text{ wins 1st, 2nd games})$

$P(A \text{ wins 1st, B wins 2nd})$

$P(A \text{ wins 1st, A wins 2nd})$

4. Solution. ~~Define~~ Denote the events

$A = \{ \text{the person is the eldest son} \}$

$B = \{ \text{the person is a boy} \}$

Obviously, $A \cap B = A$

~~$A \cap B = A$~~

$$P(A) = \frac{1}{8} \times \frac{1}{3} + \frac{3}{8} \times \frac{1}{3} + \frac{3}{8} \times \frac{1}{3} = \frac{7}{24}$$

$$P(B) = \frac{1}{2}$$

$$\Rightarrow P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)}{P(B)} = \frac{7}{12}$$